

PROBABILISTIC SYSTEMS ANALYSIS AND APPLIED PROBABILITY

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LECTURE 1: PROBABILITY MODELS AND AXIOMS

Probability gives us a mathematical way to reason about uncertainty. A **PROBABILITY MODEL** specifies what can happen and assigns probabilities to events. Its two ingredients are a sample space and a probability law.

1.1 Sample spaces: what can happen?

Before assigning probabilities, decide what one complete outcome records. The detail must be enough to answer the questions of interest.

Definition 1.1 [SAMPLE SPACE]. The **SAMPLE SPACE**, denoted by Ω , is the set of all possible outcomes of an experiment. Its outcomes must be **MUTUALLY EXCLUSIVE**: only one outcome occurs in a run. They must also be **COLLECTIVELY EXHAUSTIVE**: every possible result is included.

Example: *Two rolls of a tetrahedral die.* Each roll produces one of 1, 2, 3, 4. Write X for the first result and Y for the second. One complete outcome is the ordered pair (X, Y) , so

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ outcomes. The order matters: $(1, 2)$ and $(2, 1)$ describe different results. We can display the pairs in a grid or describe the two rolls sequentially with a tree. Each full path through the tree ends at one of the same 16 outcomes.

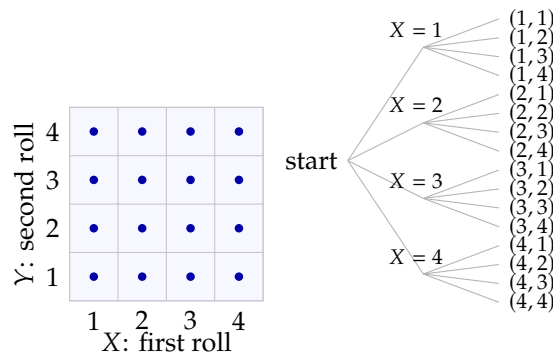


Figure 1: The grid and the tree describe the same sample space. A cell or a complete path records both rolls.

Example: *Two numbers in the unit interval.* If each result can be any real number from 0 to 1, the sample space is

$$\Omega = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\} = [0, 1]^2.$$

It is the entire unit square, including its boundary. The die space is finite; this space is continuous. Both still need a probability law.

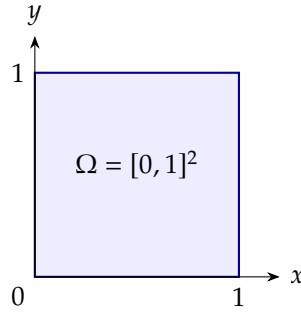


Figure 2: Each point in the unit square is one possible ordered pair.

1.2 Events and probability axioms

We often ask whether a result satisfies a condition, rather than whether one particular outcome occurs.

Definition 1.2 [EVENT AND PROBABILITY LAW]. An event A is a subset of Ω : it occurs when the outcome belongs to A . A PROBABILITY LAW assigns a number $P(A)$ to each event in its domain, subject to the probability axioms below.

For a finite or countably infinite sample space, every subset can be an event. On a continuous space, the domain of the law must be chosen with care; we use ordinary geometric regions here.

Definition 1.3 [PROBABILITY AXIOMS]. For events in the domain of P :

1. NONNEGATIVITY: $P(A) \geq 0$ for every event A .
2. NORMALIZATION: $P(\Omega) = 1$.
3. ADDITIVITY: for DISJOINT events A and B , meaning $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B).$$

The first two axioms say that probabilities cannot be negative and that some outcome in the sample space is certain to occur. The third lets us add probabilities when the events cannot occur together. For overlapping events, this sum would count the common outcomes twice.

Distinct individual outcomes give disjoint singleton events. Repeatedly applying additivity therefore gives

$$P(\{s_1, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\}).$$

We may abbreviate $P(\{s\})$ to $P(s)$. The two-event axiom handles finite unions; an infinite union will require a stronger form of additivity.

1.3 A finite probability law: two die rolls

Return to the 16 ordered die outcomes and assign probability $1/16$ to each. The probabilities are nonnegative and sum to $16(1/16) = 1$. An event containing k outcomes has probability $k/16$.

Example: *Either (1, 1) or (1, 2).* These are two distinct outcomes, so

$$P(\{(1, 1), (1, 2)\}) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

Example: *The first roll is 1.* The second roll can be 1, 2, 3, or 4. Thus

$$\{X = 1\} = \{(1, 1), (1, 2), (1, 3), (1, 4)\}, \quad P(X = 1) = \frac{4}{16} = \frac{1}{4}.$$

The notation $\{X = 1\}$ names the subset of outcomes satisfying the condition.

Example: *The sum is odd.* One roll must be odd and the other even. There are two odd values and two even values, so there are $2 \times 2 = 4$ pairs of each order. Hence

$$P(X + Y \text{ is odd}) = \frac{4 + 4}{16} = \frac{1}{2}.$$

Example: *The smaller roll is 2.* Both rolls must be at least 2, and at least one must equal 2. The event is

$$\{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

There are five outcomes, giving

$$P(\min(X, Y) = 2) = \frac{5}{16}.$$

Counting three pairs with $X = 2$ and three with $Y = 2$ would count $(2, 2)$ twice, so it must be included only once.

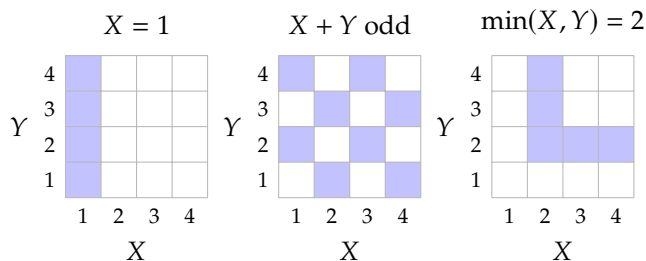


Figure 3: Highlighted cells show the last three die events. Each cell has probability $1/16$. In the last panel, the corner $(2, 2)$ is counted once.

1.4 The discrete uniform law

When a finite sample space has N equally likely outcomes, normalization forces each outcome to have probability $1/N$. This is the DISCRETE UNIFORM LAW. Additivity then gives

$$P(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ is the number of outcomes in A and $|\Omega| = N$. Computing a probability becomes a counting problem. Fair coins, fair dice, and well-shuffled decks are modeled this way. The formula relies on the outcomes being equally likely.

1.5 The continuous uniform law

Now give the unit square a CONTINUOUS UNIFORM LAW: the probability of a region is its area. The square's total area is 1, so the law is normalized. Equal-area regions have equal probabilities, and areas add for disjoint regions.

Example: *The two numbers sum to at most $1/2$.* The event is

$$A = \{(x, y) \in [0, 1]^2 : x + y \leq \tfrac{1}{2}\}.$$

It is the triangle bounded by the two coordinate axes and the line $x + y = 1/2$. Both perpendicular sides have length $1/2$, so

$$P(X + Y \leq \tfrac{1}{2}) = \text{area}(A) = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{8}.$$

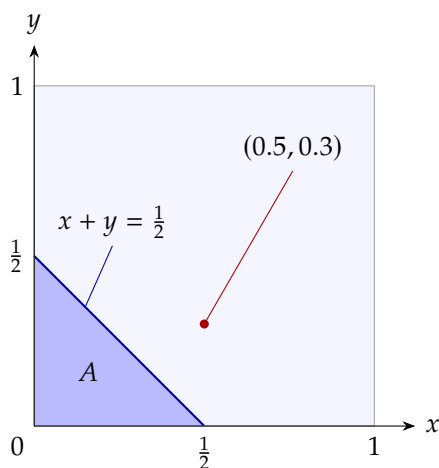


Figure 4: The shaded triangle has area $1/8$. The point $(0.5, 0.3)$ is a possible outcome, but its singleton has zero area.

Example: *The exact pair* $(0.5, 0.3)$. A single point has zero area, so

$$P((X, Y) = (0.5, 0.3)) = 0.$$

The point is still in the sample space. A probability-zero event need not be impossible: in this model every individual point has probability zero, while the whole square has probability one.

1.6 A countably infinite law and countable additivity

Consider the sample space $\Omega = \{1, 2, 3, \dots\}$ with

$$P(n) = 2^{-n}, \quad n = 1, 2, \dots$$

Each probability is positive, and the geometric series checks normalization:

$$\sum_{n=1}^{\infty} 2^{-n} = \frac{1/2}{1 - 1/2} = 1.$$

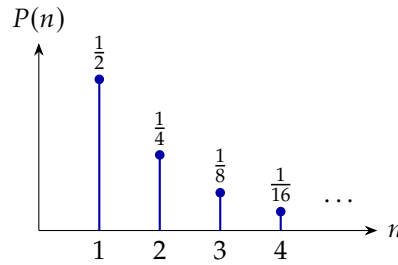


Figure 5: The first four outcome probabilities. Each is half the preceding one, and the pattern continues for every positive integer.

Example: *The outcome is even.* The event $\{2, 4, 6, \dots\}$ contains infinitely many disjoint singleton events. Adding their probabilities gives

$$\begin{aligned} P(\{2, 4, 6, \dots\}) &= \sum_{k=1}^{\infty} 2^{-2k} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \\ &= \frac{1/4}{1 - 1/4} = \frac{1}{3}. \end{aligned}$$

This calculation uses an infinite sum, which finite additivity alone does not justify.

Definition 1.4 [COUNTABLE ADDITIVITY]. If $A_1, A_2, \dots$ are pairwise disjoint events, the COUNTABLE ADDITIVITY axiom requires

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

This is the full additivity axiom for a probability law. It covers a sequence indexed by the positive integers and includes finite additivity as a special case. It justifies the even-outcome calculation above. It does not permit summing singleton probabilities over all the points of the unit square: that collection is uncountable.

Revision. A model needs both a sample space and a probability law. Events are sets of outcomes. Count outcomes under a finite uniform law, measure area under the uniform law on the unit square, and use countable additivity for a countable union of disjoint events.

Common mistakes: using the counting formula without equal likelihood, counting a shared outcome twice, or treating probability zero as impossibility.

Check yourself: Why do two tetrahedral-die rolls give 16 outcomes? Why does $\min(X, Y) = 2$ select five of them? Why can an infinite sum give the probability of an even integer, but cannot add up the probabilities of every point in the unit square?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 1: Probability Models and Axioms](#), PDF pp. 1–3. The calculations are completed and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).